Effect Fidelity: can an AI simulator replace an A/B test?

I ran small AI persona simulators on 17 real randomized experiments (Many Labs 1, GSS 2018 wording experiments, the Hillstrom email test) and checked the ship / no-ship call each one would have led to.

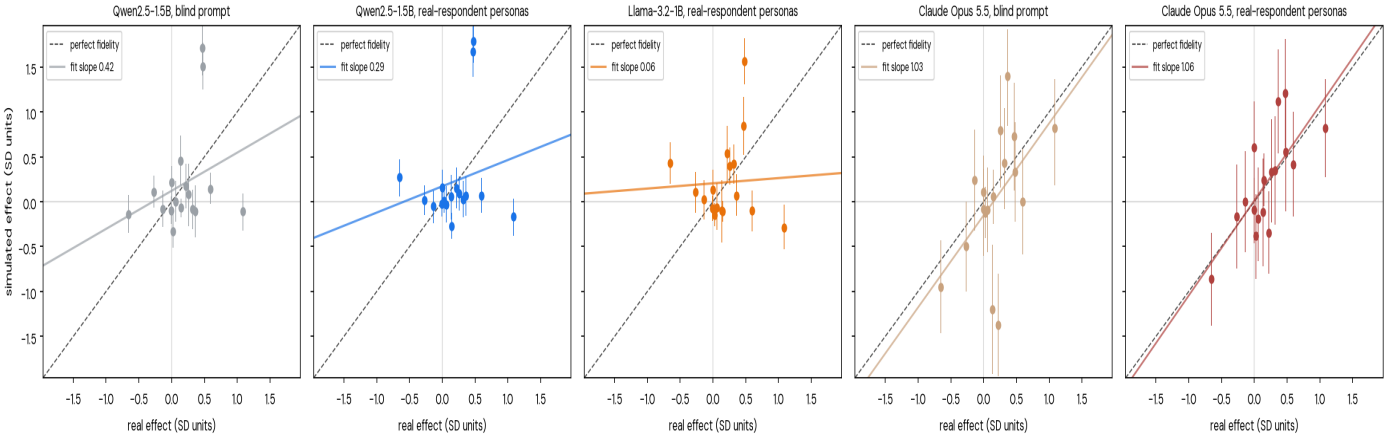
Small open models' simulated effects barely track the real ones (correlation 0.05 to 0.19); one quick run of Claude Opus 5.5 tracks them well (0.73). But at 25 simulated people per arm every simulator still made the wrong ship / keep / no-difference call in 47% to 64% of experiments, and mixing simulators with real respondents added about nothing.

How much to trust the wrong-call number

A call is ship B, keep A, or no clear difference, read from a 95% interval; it is wrong if it differs from what the full real study supports. With these 17 experiments, always shipping B would be wrong 41% of the time and always saying 'no difference' 76%, so the rate is weak evidence on its own. A simulator's interval also narrows as you run more simulated people, so all simulators are compared at 25 per arm. Effect correlation and direction agreement separate them better. Full audit: audit/AUDIT.md.

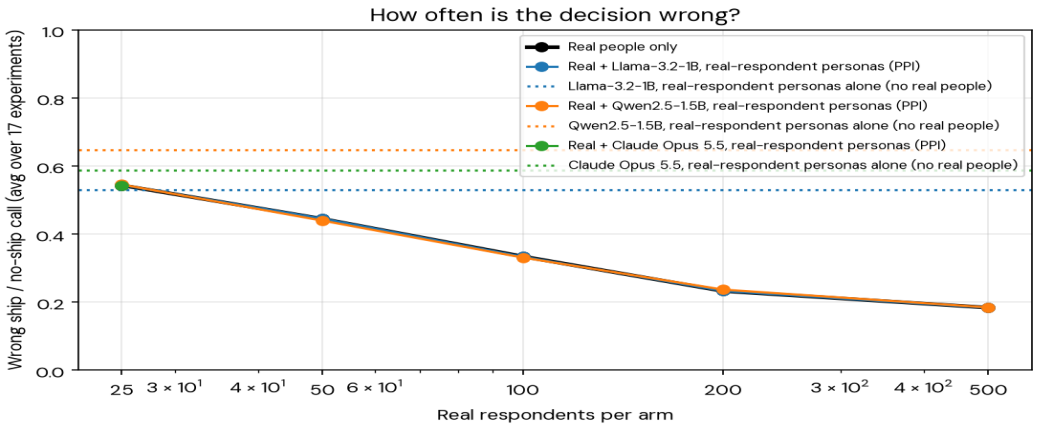
1. Realistic answers, wrong effects

In 23 simulated experiments both arms matched the real answer distributions closely (TVD under 0.15). 13 of those still gave the wrong call. The small models shrank effects toward zero (calibration slopes 0.06 to 0.42, where 1.0 is right). Their 95% intervals held the real effect in only 29% to 76% of experiments. Li & Ji (2026) report the same gap.



2. Real people plus a simulator: worth about zero extra people

Prediction-powered inference (the mixed subjects design of Broska et al. 2025) lets a simulator stretch a small real sample. Here the gain was 1.00x at 100 real people per arm, so the simulator was worth almost no one. The reason: each simulated person's answer had almost no relation to the real person's answer (median correlation within ±0.14). Real people alone, 100 per arm, still got the call wrong about 33% of the time, and adding the simulator didn't change that. The method stayed honest: it fell back to the real data.



3. Fixes that help the numbers but not the decisions

A small LoRA fine-tune of Qwen2.5-1.5B on SocSci210 (Kolluri, Wu, Park & Bernstein 2025) cut effect-size error by 12%. Wrong calls did not improve (65% before, 71% after). A held-out correction (one scale factor plus a model-error term) brought interval coverage to 88% to 94%. It does that by widening the intervals, so it doesn't fix the calls either.

4. A frontier model: Claude Opus 5.5

A quick Claude Opus 5.5 run (25 simulated people per arm, through the Claude Code CLI) got the size of effects about right (calibration slope 1.06) and cut effect error by 32% versus the small persona models. It still made the wrong call (raw, bootstrap interval) in 53% of experiments with personas and 47% with a blind prompt. Mixed with real people (20 per arm, the most this quick run's 40-person pool allows) it was worth 1.03x, again about nothing.

Next

A full-size Claude run (more simulated and real people per arm), and finding which kinds of questions a simulator can be trusted on. The harness works with any model.

Caveats: 1 to 1.5B-parameter models on CPU plus one quick Claude run, 17 experiments, one fine-tuning run on about 480 examples, text descriptions in place of visual primes. Code and data: github.com/tommuthecow/effect-fidelity. Li & Ji (2026), arXiv 2604.02458. Broska, Howes & van Loon (2025), doi:10.1177/00491241251326865. Kolluri, Wu, Park & Bernstein (2025), arXiv 2509.05830.